

Youngjun Park

PostDoc Researcher, School of Computing, Korea Advanced Institute of Science and Technology

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Research Interests

As an urban researcher with a background in Architecture and Urban Design, my academic journey has focused on exploring the relationship between human activities and urban environments. My doctoral work, centered on walking activities in urban parks, laid the groundwork for my evidence-based approach to urban studies, providing initial insights into analyzing human behavior in urban contexts.

Currently, as a postdoctoral researcher in Computer Science, I have expanded my expertise to integrate scientific and mathematical frameworks into the analysis of urban planning and design issues. This role has also honed my abilities in managing research projects and leading cross-disciplinary collaborations, drawing on my experience as a lecturer and my trained leadership as a former military officer.

With a profound commitment to evidence-based urban design and unique experience in urban design, public health, and data science, I am eager to contribute to research that supports the idea of healthy and sustainable cities.

Education

Ph.D. Architecture, Seoul National University, 2022

M.S. Urban Design, Seoul National University, 2017

B.A. Architecture, Seoul National University, 2013

Professional Positions

2024-present **Partner Architect.** Ar-ge Architects, Inc.

2022-present **PostDoc Researcher.** School of Computing, Korea Advanced Institute of Science and Technology (KAIST)

2021-2023 **Lecturer.** Department of Architecture, Seoul National University (SNU)

2013-2015 **Military Officer (1st Lieutenant).** Combat Engineer, Republic of Korea Army.

Publications

Journal Article (Peer-reviewed)

- 1 Park, Youngjun; Chung, Haegwon; Park, Sohyun (2021) Changes of walking activity during the first cycle phases of COVID-19 pandemic: A case study of Seoul, Korea. *HERD: Health Environments Research & Design Journal*, 14(4), 43-57.
- 2 Park, Youngjun; Lee, Sunjae; Park, Sohyun (2022) Differences in park walking, comparing the physically inactive and active groups: data from mHealth monitoring system in Seoul. *International journal of environmental research and public health*, 19(1), 395.
- 3 Jeong, Jiwoon; Park, Youngjun; Park, Sohyun (2023) Safety-critical events in bicycle lanes in Jongno, Seoul. *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 16(1), 1-24.

- 4 Park, Youngjun; Park, Sohyun (2019) Car-use and Non-work Travel of Housewives in Residential Neighborhood of Seoul. *Seoul Research*, 20(1), 45-59.

Conference Proceedings & Presentations

- 1 Park, Youngjun; An, Jisun; Lee, Dongman (2025) A Model for Monthly, Local-Level Airbnb Changes Using Public Dataset (academic paper), *Proceedings of the 8th International Conference on Tourism Research*, 8(1), 214-223.
- 2 Park, Youngjun; Han, Sumin (2024) Enhancing Population Predictions in Developing Area through Building Construction Data (poster session), *Association of European Schools of Planning (AESOP'24)*
- 3 Han, Sumin; Park, Youngjun; Sonia Sabir; An, Jisun; Lee, Dongman (2024) Improving Real Estate Appraisal with POI Integration and Areal Embedding. *DMO-FinTech'24: Workshop on Decision Making and Optimization in Financial Technologies (PAKDD'24)*
- 4 Park, Youngjun; Han, Sumin; Park, Doyun; Bae, Hyeongchan; Nguyen Thanh Tung; Lee, Dongman (2024-06-26) Integrative Geospatial Visualization for Urban Mobility Analysis. *Korea Computer Congress (KCC'24)*
- 5 Ahmad Elmoursi; Han, Sumin; Bae, Hyungchan; Park, Youngjun; Nguyen Thanh Tung, Lee, Dongman. (2024-06-26). Enhancing Geospatial Recommendations through Natural Language Processing with Large Language Models. *Korea Computer Congress (KCC'24)*
- 6 Han, Sumin; Park, Youngjun; Lee, Minji; An, Jisun; Lee, Dongman (2023) Enhancing Spatio-temporal Traffic Prediction through Urban Human Activity Analysis. *Information and Knowledge Management, Proceedings*, 689-698
- 7 Park, Youngjun; Han, Sumin; An, Jisun; Lee, Dongman (2023) Encoding Urban Trajectory as a Language: Deep Learning Insights for Human Mobility Pattern. *Association of Collegiate Schools of Planning (ACSP '23)*
- 8 Park, Youngjun (2022) Parks and Healthy Walking: smartphone physical activity recognition data for place and walking. *Urban Design Institute of Korea*
- 9 Park, Youngjun; Park, Sohyun (2017) A Study on the Location of Car-sharing in the Neighborhood Area and Replacement of Vehicle Ownership in Community. *Urban Design Institute of Korea*

Patents & Copyrights

- 1 Korean Patent (applied) No.10-2024-0108083 Method and Device for Estimating House Value Using Neural Network Model
- 2 Korean Patent (registered) No.10-23679770000 Method and Server for Providing Commercial Real Estate Information
- 3 Korean Copyright (registered) C-2023-062275 Geographic Data Visualization Program

Research Projects

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| 2024-current | PostDoc Researcher
Center for Advanced Urban System (G01240393)
Korea Advanced Institute of Science and Technology (KAIST), GS Engineering & Construction Corporation |
| 2024-current | PostDoc Researcher
AI-based Contactless Drug Detection Technology (RS-2024-00459749)
Korea Advanced Institute of Science and Technology (KAIST) |
| 2024-current | PostDoc Researcher
Multimodal Spatiotemporal Urban Prediction Model Based on Large-Language Model (RS-2024-00356597) |

2022-current	Korea Advanced Institute of Science and Technology (KAIST) PostDoc Researcher Self-learning Based Autonomic IoT Edge Computing Technology SW StarLab Project (RS-2019-II191126)
2022-2024	Korea Advanced Institute of Science and Technology (KAIST) Participatory Researcher AI-based Smart Transportation System, Daegu city (N04230030)
2022-2023	Korea Advanced Institute of Science and Technology (KAIST) Research Consultant Text analysis of the Government Budget for Pedestrian Projects Architecture and Urban Research Institute of Korea (AURI)
2019-2020	Leading Researcher Data analysis, mobile app design & development for WalkON™ Swallaby, Inc.
2018-2019	Research Assistant GIS analysis of Urban Environment for Infant Welfare Architecture, Ewha Womans University
2015-2016	Research Intern Urban design and masterplan, Hue city, Vietnam (ODA project of KOICA) Graduate School of Environmental Studies, Seoul National University

Lectures & Courses

Courses at Seoul National University (SNU), Architecture and Urban Design

- “Urban Design Analysis: The Pedestrian and the City”, for Master students in the department of Architecture (Spring 2021, 3 credits)
- “Theory of Urban Design: From Classic to Today”, for Master students in the graduate program of Urban Design (Spring 2021, 3 credits)

Invited Lectures

- “Walking, Physical Health, and Urban Environment”, for Master students in the department of Architecture, Seoul National University (SNU) (Spring 2022)
- “Smart City and Urban Data”, for Undergraduate students in the department of Architecture, Korea University of Technology and Education (Spring 2024)
- “Ubiquitous Social Computing”, for Graduate students in the School of Computing, Korea Advanced Institute of Science and Technology (KAIST) (Fall 2024)
- “Smart City and Sustainable City”, for Undergraduate students in the department of Architecture, Chungang University (Spring 2025)

Awards

2023.11	G-School Best Innovation Awards, “Spatial Awareness in Deep Learning: Approaches to Integrating Geolocational Attributes in Deep Learning Models”, KAIST-NYU Young Researcher Day, Urban X, KAIST & NYU (Video Link)
2023.10	1 st prize, “AI-based Population Prediction”, Data Analysis Competition, Korea Land & Housing Corporation
2018.04	Thesis Award (Master thesis), Architectural Institute of Korea
2016.03	Young Architectural Fellowship, Korean Institute of Architects
2015.01	2024 Excellent Officer Award, Republic of Korea Army